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Predicting Protein Crystallization Conditions

Data Acquisition, Preprocessing, Exploratory Data Analysis

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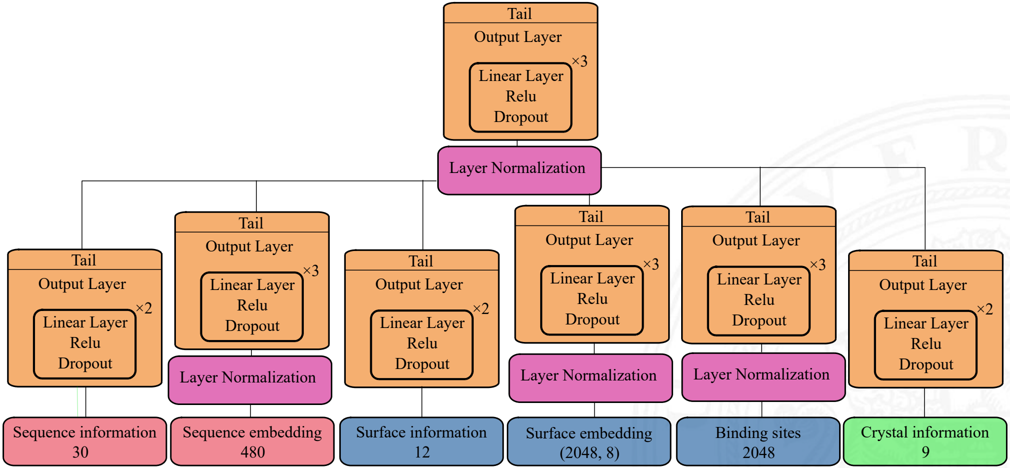
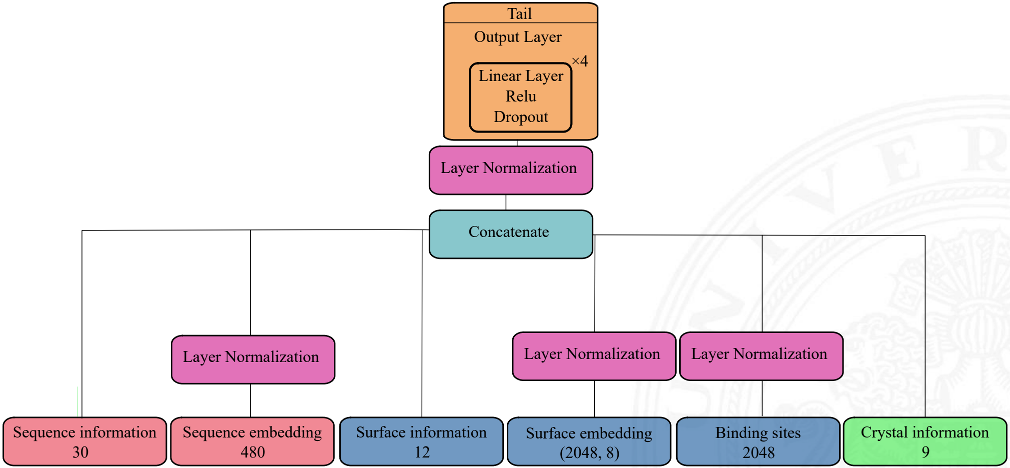
Model Architecture Separate Tails

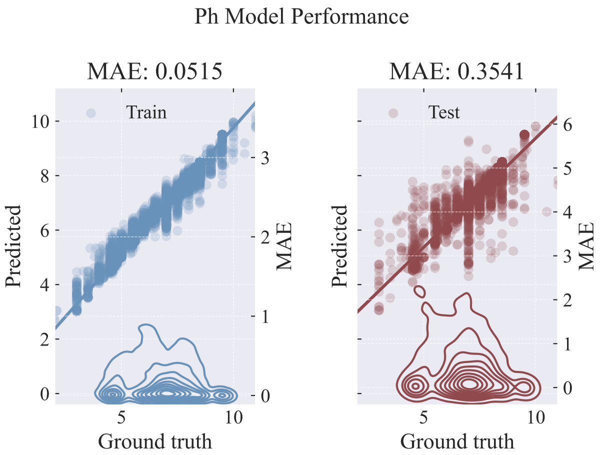


Model Architecture Joined Tail



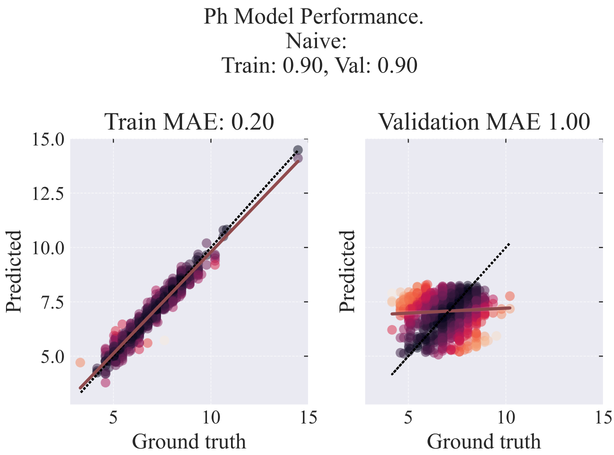
Performance on Ph Prediction task

- Validation split 0.2 based on unique entry id similar to Lee et al. (2019)
- Explained Variance:
  - Train: 0.96
  - Validation: 0.89
- Pearson:
  - Train: 0.98
  - Validation: 0.9

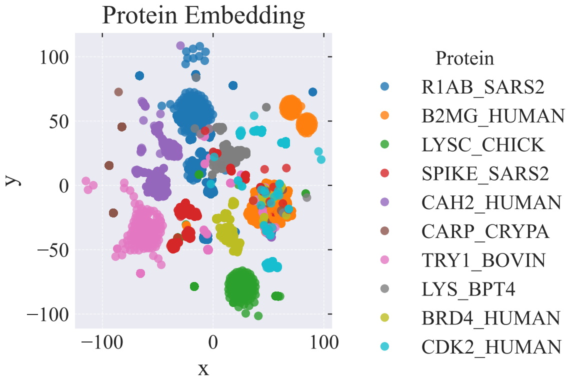


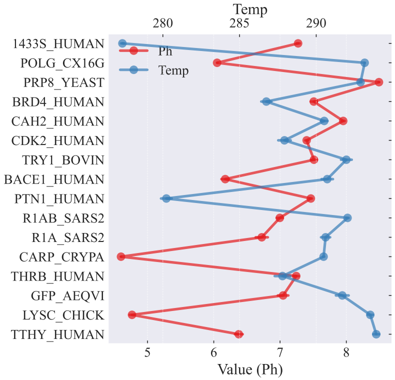
Performance on Ph Prediction task with valid split

- protein based validation split instead of entry based
- build bi-partition tree to avoid data leakage from files with multiple proteins
- Explained Variance:
  - Train: 0.96
  - Validation: 0.25
- Pearson:
  - Train: 0.98
  - Validation: 0.5



Performance on Ph Prediction task with valid split

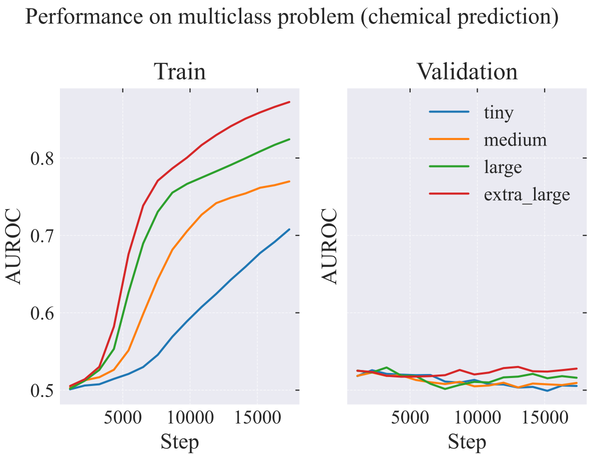




(a) Median and variance for Ph/Temperature for a given protein

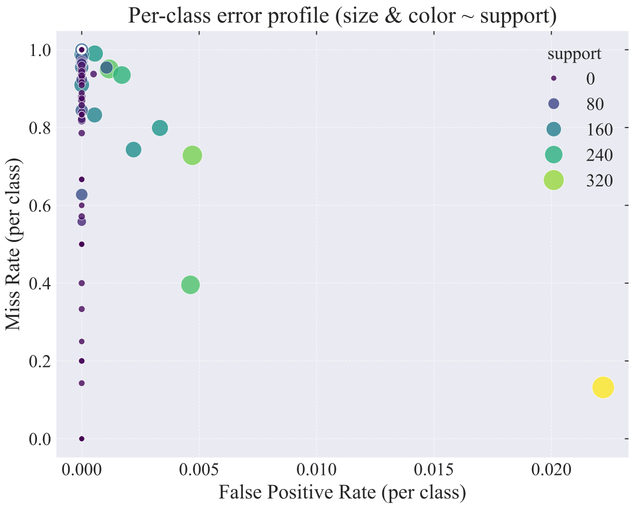
Prediction of chemicals

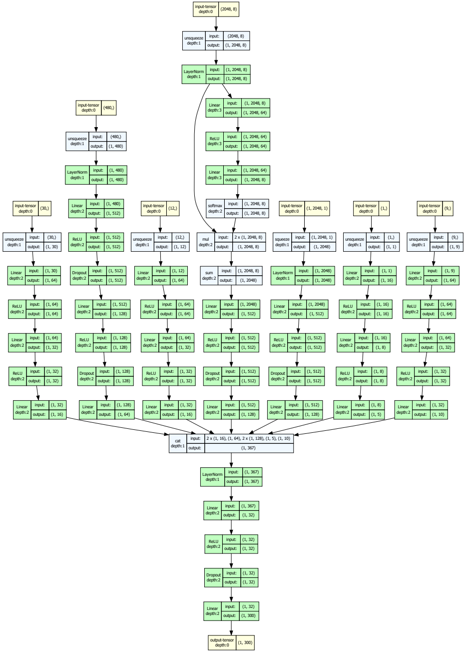
- multi-hot encode chemicals ([0, 0, 1, 1, 0, ..., 0, 1, 0])
- top 300 chemicals
- for inference check for top



Prediction of chemicals

- Even for training the predictions are pretty bad
- Model performance strongly biased towards often occurring chemicals





input-tensor depth:0 (2048, 8)
unsqueeze depth:1 input: (2048, 8) output: (1, 2048, 8)
LayerNorm depth:1 input: (1, 2048, 8) output: (1, 2048, 8)
Linear depth:3 input: (1, 2048, 8) output: (1, 2048, 64)
ReLU depth:3 input: (1, 2048, 64) output: (1, 2048, 64)
Linear depth:3 input: (1, 2048, 64) output: (1, 2048, 8)
softmax depth:2 input: (1, 2048, 8) output: (1, 2048, 8)
mul depth:2 input: 2 x (1, 2048, 8) output: (1, 2048, 8)
sum depth:2 input: (1, 2048, 8) output: (1, 2048)
Linear depth:2 input: (1, 2048) output: (1, 512)
ReLU depth:2 input: (1, 512) output: (1, 512)
Dropout depth:2 input: (1, 512) output: (1, 512)
Linear depth:2 input: (1, 512) output: (1, 128)
input-tensor depth:0 (480,)
unsqueeze depth:1 input: (480,) output: (1, 480)
LayerNorm depth:1 input: (1, 480) output: (1, 480)
Linear depth:2 input: (1, 480) output: (1, 512)
ReLU depth:2 input: (1, 512) output: (1, 512)
Dropout depth:2 input: (1, 512) output: (1, 512)
Linear depth:2 input: (1, 512) output: (1, 128)
ReLU depth:2 input: (1, 128) output: (1, 128)
Dropout depth:2 input: (1, 128) output: (1, 128)
Linear depth:2 input: (1, 128) output: (1, 64)
input-tensor depth:0 (30,)
unsqueeze depth:1 input: (30,) output: (1, 30)
Linear depth:2 input: (1, 30) output: (1, 64)
ReLU depth:2 input: (1, 64) output: (1, 64)
Linear depth:2 input: (1, 64) output: (1, 32)
ReLU depth:2 input: (1, 32) output: (1, 32)
Linear depth:2 input: (1, 32) output: (1, 16)
input-tensor depth:0 (12,)
unsqueeze depth:1 input: (12,) output: (1, 12)
Linear depth:2 input: (1, 12) output: (1, 64)
ReLU depth:2 input: (1, 64) output: (1, 64)
Linear depth:2 input: (1, 64) output: (1, 32)
ReLU depth:2 input: (1, 32) output: (1, 32)
Linear depth:2 input: (1, 32) output: (1, 16)
input-tensor depth:0 (1, 2048, 1)
squeeze depth:1 input: (1, 2048, 1) output: (1, 2048)
LayerNorm depth:1 input: (1, 2048) output: (1, 2048)
Linear depth:2 input: (1, 2048) output: (1, 512)
ReLU depth:2 input: (1, 512) output: (1, 512)
Dropout depth:2 input: (1, 512) output: (1, 512)
Linear depth:2 input: (1, 512) output: (1, 128)
input-tensor depth:0 (1,)
unsqueeze depth:1 input: (1,) output: (1, 1)
Linear depth:2 input: (1, 1) output: (1, 16)
ReLU depth:2 input: (1, 16) output: (1, 16)
Linear depth:2 input: (1, 16) output: (1, 8)
ReLU depth:2 input: (1, 8) output: (1, 8)
Linear depth:2 input: (1, 8) output: (1, 5)
input-tensor depth:0 (9,)
unsqueeze depth:1 input: (9,) output: (1, 9)
Linear depth:2 input: (1, 9) output: (1, 64)
ReLU depth:2 input: (1, 64) output: (1, 64)
Linear depth:2 input: (1, 64) output: (1, 32)
ReLU depth:2 input: (1, 32) output: (1, 32)
Linear depth:2 input: (1, 32) output: (1, 10)
cat depth:1 input: 2 x (1, 16), (1, 64), 2 x (1, 128), (1, 5), (1, 10) output: (1, 367)
LayerNorm depth:1 input: (1, 367) output: (1, 367)
Linear depth:2 input: (1, 367) output: (1, 32)
ReLU depth:2 input: (1, 32) output: (1, 32)
Dropout depth:2 input: (1, 32) output: (1, 32)
Linear depth:2 input: (1, 32) output: (1, 300)
output-tensor depth:0 (1, 300)

Sources

Lee, Hyunim et al. (2019). "De novo Crystallization Condition Prediction with Deep Learning". In: *Advances in Neural Information Processing Systems*. Vol. 32. Accessed: 2025-07-31. URL: https://mlcb.github.io/mlcb2019_proceedings/papers/paper_3.pdf.